

# KATRIEL CALEB O. TIRU

Software Engineer · Full-Stack Development

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## PROFESSIONAL SUMMARY

Computer Science graduate who builds full-stack web applications end to end: React and Next.js front ends, Node.js and Express services, background job pipelines, and PostgreSQL on AWS and Supabase. Comfortable in Agile, CI/CD-driven teams and known for shipping clean, well-tested code. Seeking a software engineering role where I can deliver production features and keep growing as a reliable developer.

## EDUCATION

### Bachelor of Science in Computer Science

Holy Angel University · Angeles, Pampanga

Graduated April 2026

## PROFESSIONAL EXPERIENCE

### AI Model Evaluator (Independent Contractor)

DataAnnotation

Jun 2026 – Present

Remote

- Run **head-to-head evaluations** of frontier **large language models** from multiple providers on unfamiliar public repositories chosen for real-world complexity.
- Judge which of two model outputs is stronger against structured **rubrics** for correctness, instruction-following, and developer experience, writing the assessments that rank model performance.

### Software Engineer Intern

TwistResources Inc.

Jan 2026 – May 2026

Angeles, Pampanga

- Delivered features for **TwistHRM** across two Agile sprints, using AI-assisted tooling to produce review-approved code that integrated cleanly with the existing **CI/CD pipeline**.
- Engineered an **automated data import service** that normalized external CSV formats (KAMI, Sprout) into the platform schema, removing manual re-entry from the onboarding workflow.
- Built an **image processing utility** that optimized UI assets for faster page loads while preserving original file integrity.

## PROJECTS

### Aquarium Monitoring Application (Undergraduate Thesis)

2025 – 2026

- Developed a real-time monitoring system on **Raspberry Pi** combining a **CNN-based fish behavior recognition** model with live water-quality sensor data.
- Implemented automated threshold alerts and a mobile dashboard so owners could act on abnormal readings or behavior as they happened.

### UptimeKit: Uptime & Performance Monitoring Platform

2026

Next.js 14 · TypeScript · BullMQ / Redis · Docker · Supabase PostgreSQL

[github.com/kcotiru/UptimeKit](https://github.com/kcotiru/UptimeKit)

- Built a self-hosted Pingdom alternative: a **BullMQ worker** checks endpoints every 30 seconds, logs outages with duration and cause, and alerts **Slack, Discord, or webhooks**.
- Kept p50/p95/p99 latency charts fast and storage flat with a **three-tier rollup pipeline** (raw → hourly → daily), and isolated tenants in **PostgreSQL row-level security**, with SSRF blocking and **Vitest** integration tests.

### REST URL: URL Shortening Service

2026

Node.js · Express · TypeScript · React / Vite · Supabase PostgreSQL

[github.com/kcotiru/REST-URL](https://github.com/kcotiru/REST-URL)

- Built a full-stack URL shortener with auto-generated **nanoid** codes, user-defined custom slugs, and atomic access tracking, served by a layered **REST API** (controller → service → repository) with **Zod** request validation.

## TECHNICAL SKILLS

### Languages

TypeScript, JavaScript, SQL

### Front-End

React, Next.js, Tailwind CSS, HTML5, CSS3

### Back-End

Node.js, Express, REST API design, background job queues (BullMQ, Redis)

### Databases

PostgreSQL, Supabase

### Cloud & DevOps

AWS EC2, Amazon S3, AWS CodeBuild, AWS CodeCommit, Docker, CI/CD pipelines

### Practices & Tools

Agile / Scrum, Git & GitHub, AI-assisted development